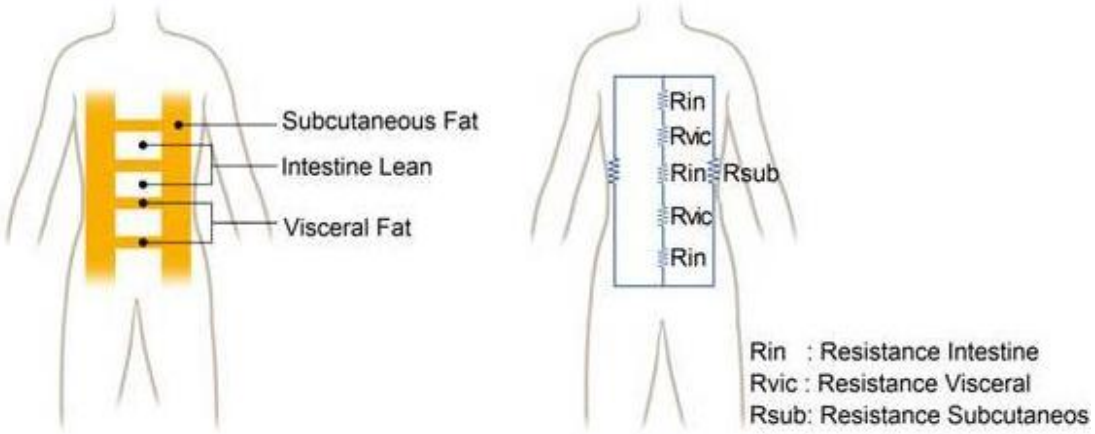
Principle of calculating visceral fat area from InBody

→ What are the elements used for calculating visceral fat?

The visceral fat area is not directly measured by bioelectrical impedance measurement (BIA). However, the possibility of calculating visceral fat area through BIA method was suggested from the papers which studied the relationship between trunk impedance and visceral fat. [1),2)]

Based on this theoretical background, the visceral fat area provided by InBody is calculated from the regression formula of the visceral fat area taken by the CT scans. While doing this, **BMI, fat free mass, body fat mass** and **percent body fat**, and **segmental lean muscle** distribution are considered as well as the total body impedance obtained by the segmental measurement.

In the case of the trunk, it consists of subcutaneous fat (intestinal fat), intestine lean (visceral muscle), visceral fat (visceral fat) accumulated in the subcutaneous tissue as shown below.



Subcutaneous fat is present in the subcutaneous tissue and *Rsub* is connected in parallel to the current flowing vertically through the trunk. On the other hand, Intestine Lean and Visceral Fat can be seen to have a series of connection of *Rin* and *Rvic* for the current flowing vertically through the trunk.

Therefore, if Intestine Lean and Visceral Fat are unchanged and the subcutaneous fat increases, the outer wall of the fat thickens. However, electric current flows along the intestinal muscles, so the effect on the impedance is less. On the other hand, if there is no change in the subcutaneous fat and only Visceral Fat is increased, the series of connected *Rvic* increases and directly affects the impedance of the body, therefore it changes the size and the characteristics of the impedance.

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1) Detection of segmental internal fat by bioelectrical impedance analysis in a biological phantom. He Q, Wang J, Engelson ES, Kotler DP. Nutrition 19(6):541-544, 2003

2) A new simple method for the measurement of visceral fat accumulation by bioelectrical impedance. Diabete Care 28(2):451-3, 2005

Using these characteristics of the impedance, the regression equation which estimates the visceral fat area of the abdomen (Umbilicus line) measured by the CT scan can have a high correlation around R = 0.8. For the correlation between the visceral fat area obtained from CT and the visceral fat area obtained from InBody, please refer to the following article. [3) 4)]

InBody 720 as a New Method of Evaluating Visceral Obesity

Hisataka Ogawa[1], Kazumasa Fujitani[1], Toshimasa Tsujinaka[1], Kenji Imanishi[2], Hatsumi Shirakata[2], Aiko Kantani[2], Motohiro Hirao[1], Yukinori Kurokawa[1], Shigetoshi Utsumi[2]
[1]Department of Surgery and [2]Nutrition Support Service,
National Osaka Medical Center, Osaka, Japan.

ABSTRACT

Background/Aims: This study was conducted to assess the efficacy of bioelectrical impedance analysis by InBody 720 as a new tool for measuring visceral fat area.
Methodology: A total of 53 consecutive patients elected to undergo surgical resection of primary gastric cancer received preoperative measurement of visceral fat area at the umbilical level with both computed tomography and bioelectrical impedance analysis by InBody 720.
Results: Visceral fat area values measured by InBody 720 significantly correlated with those by computed tomography (R=0.759).
Conclusion: Bioelectrical impedance analysis by InBody 720 was shown to be useful as a more convenient substitute for computed tomography when measuring visceral fat area.

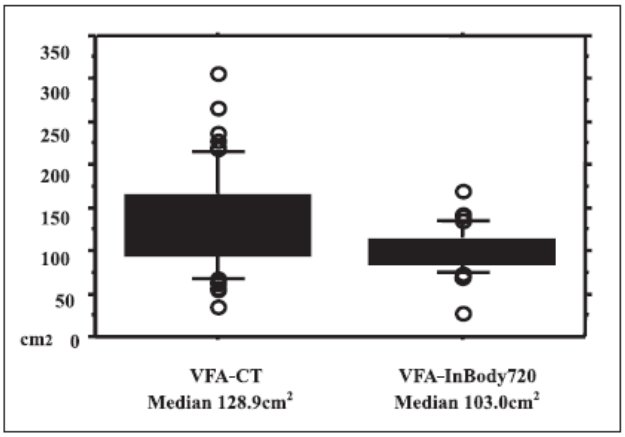


FIGURE 2 Box–and–whisker plots of VFA measured by CT scan and InBody 720.

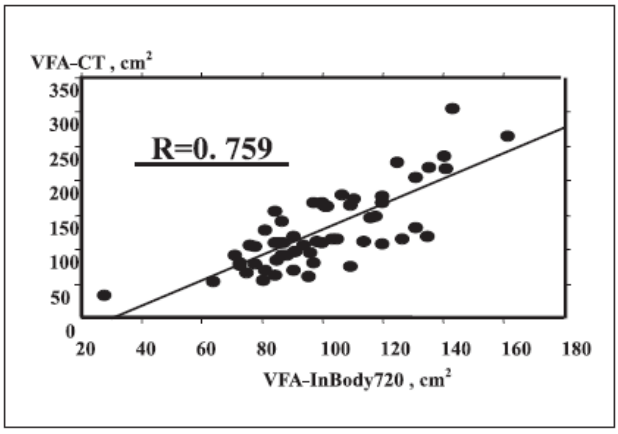


FIGURE 3 A scatter plot of VFA.

3) Hisataka Ogawa et al, InBody720 as a new method of evaluating visceral obesity. Hepato-Gastroenterology. 58:42-44, 2011

The Effects of Androgen Deprivation Therapy on Lipid Metabolism and Body Composition in Japanese Patients with Prostate Cancer

Kazumasa Torimoto[1,*], Shoji Samma[2], Yoriaki Kagebayashi[2], Yoshitomo Chihara[3], Nobumichi Tanaka[1], Akihide Hirayama[1], Kiyohide Fujimoto[1] and Yoshihiko Hirao[1]

[1]Department of Urology, Nara Medical University, Kashihara, [2]Department of Urology, Nara Prefectural Nara Hospital, Nara and [3]Department of Molecular Pathology, Nara Medical University, Kashihara, Japan

Objective: In Japan, androgen deprivation therapy is employed as the primary therapy for prostate cancer in more than 50% of patients, which is a percentage larger than that in the USA. The adverse effects of androgen deprivation therapy on body composition and lipid profile associated with metabolic syndrome have been reported mainly in Caucasian populations, and few studies have been performed in East Asian populations, including Japanese.

Methods: This study enrolled 39 Japanese patients who were starting to receive androgen deprivation therapy for prostate cancer. Subjects were evaluated at baseline and at 3, 6, 9 and 12 months. Body composition and lipid profiles were measured by bioelectrical impedance analysis and using blood samples, respectively.

Results: The volume of fat and visceral fat was significantly increased 6 months after the treatment and continued to increase until 12 months. On the other hand, skeletal muscle was significantly decreased during the same period. The serum concentration of total cholesterol and low-density lipoprotein cholesterol increased significantly over the same period.

Conclusions: Androgen deprivation therapy changed the body composition and lipid profile of men with prostate cancer. It was demonstrated that even Japanese patients with prostate cancer who are treated with androgen deprivation therapy have the risk of developing metabolic syndrome.

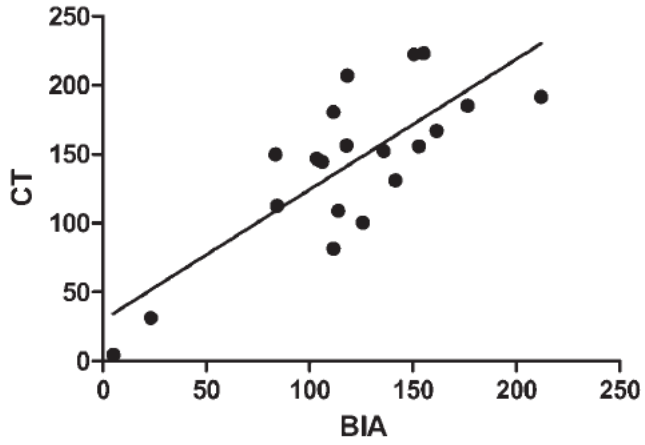


Figure 1. Visceral fat area (VFA) measured by bioelectrical impedance analysis significantly correlated with VFA measured at computed tomography in 20 patients ($r^2 = 0.6782$, $P < 0.001$).

4) Kazumasa Torimoto et al, The Effects of Androgen Deprivation Therapy on Lipid Metabolism and Body Composition in Japanese Patients with Prostate Cancer. Jpn J Clin Oncol. 41(4)577–581, 2011